



Risk-Based Method for Determining Separation Minima in Unmanned Aircraft Systems

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Risk assessment is a key issue in enabling the use of unmanned aircraft systems (UASs) in nonsegregated areas, especially in very low-level airspace and urban areas. The specific operations risk assessment (SORA) methodology represents an important milestone in performing the risk assessments required by aviation safety agencies to UAS operators in specific operations. However, the SORA is a qualitative method used for UASs operating inside a well-bounded operational volume. This paper proposes a quantitative method that can not only be used in a closed volume but also in an airspace shared by several UAS missions and even general aviation. The basis for this is providing a separation volume (a “bubble”) to prevent collisions that is calculated using a risk-based approach. The method consists of setting a target level of safety, which is accomplished using a tradeoff between strategic and tactical mitigations. A probabilistic methodology for performing quantitative risk assessment of strategic and tactical mitigations is provided, and the dependence of the separation distance is carefully analyzed. All factors affecting the separation distance are identified, and their contributions to collision risk are probabilistically estimated. The method takes into account specific factors relating to UASs such as trajectories, separation methods, and performance. As a result, the method allows numerical determination of a separation distance for a given target level of safety and an operational scenario.

I. Introduction

UNMANNED aircraft system (UAS) operations have recently experienced significant growth, particularly in urban areas. This introduces new risks for people and infrastructure on the ground and for manned aircraft flying at low altitudes. The primary approach to limiting the risks of UAS operations has generally been flying in segregated areas; however, this greatly restricts UAS operations. Current UAS applications mostly seek to fly in very low-level airspace, below 500 ft (152 m). This requires sharing of the airspace with other UAS users and probably with general aviation flying with visual flight rules (VFRs). The risk is especially critical in the CTR of an airport, which often extend over urban areas with their low limit touching the ground.

New regulatory frameworks are being developed by aviation safety agencies to enable UAS operations in different airspaces while keeping risks at an acceptable level (see Refs. [1,2] for examples). These regulations define the operational concepts, airspace classes, and airspace services needed to support UAS missions. A number of airspace services to support UAS operations to be provided by UAS service suppliers are being identified according to the technical capability levels required by increasingly complex operations. The CORUS project [3] has presented a prospective classification of such services in the European context, known as U-space services, but there are researchers working on similar proposals for UASs in most countries, such as the UAS traffic management (UTM) concept of operations of the Federal Aviation Administration in the United States.^{‡‡}

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^{‡‡}“Concept of Operations v2.0,” White Paper, Federal Aviation Administration, March 2020, <https://www.ecfr.gov/current/title-14/chapter-I/subchapter-F/part-107> [retrieved 3 October 2022].

This paper focuses on assessing UAS collision risks with other aircraft, including other UASs and manned aircraft. UAS collision risks can be mitigated in two different ways: strategic and tactical. Strategic mitigation focuses on advanced planning and operational restrictions to avoid traffic conflicts. The most restrictive measure in the strategic approach is airspace segregation. Tactical mitigations try to reduce the midair collision (MAC) risk using two alternative approaches: self-separation or a tactical separation service (TSS). The latter is similar to air traffic control (ATC) and is provided by a UAS service supplier. Self-separation either requires a visual line of sight to identify potential intruders or the use of an onboard detect-and-avoid (DAA) device to identify them when visual detection is not possible. There are different views in the United States and Europe regarding the provision of a TSS for UAS: the U.S. concept of operations (ConOps) (see footnote ‡‡) states that “UTM Operators are ultimately responsible for maintaining separation,” and so the TSS only has a supporting role with no separation responsibility; conversely, the European ConOps [3] identifies the need for an advanced (U3) U-space service where the TSS has the separation responsibility in some airspaces (Zu airspace) or where the service should interact with ATC (Za airspace).

Assessing UAS operation risks is vital for authorizing so-called UAS “specific operations” in Europe. The Joint Authorities for Rule-making on Unmanned Systems group has published a methodology called the specific operations risk assessment (SORA) [4] to provide clear guidelines for this process. The SORA focuses on assigning two classes of risk to UAS operations: a ground-risk class (GRC) and an air-risk class (ARC). The GRC and ARC form the basis for determining the specific assurance and integrity levels (SAILs) for both risk classes. The SAIL represents the level of confidence that a UAS operation will stay under control within the boundaries of its intended operation. The final step in the risk assessment is the recommendation of operational safety objectives (OSOs) to be met in accordance with the SAILs.

This paper concentrates only on the ARC and UAS collisions, and it differs from the SORA in a number of ways. First, it does not assume the SORA semantic model based on enclosing the UAS inside an operational volume extended with a contingency volume. The SORA sets the maximum probability of leaving this volume to 1×10^{-4} per flight hour. In our model, the operation volumes of several UAS operations may overlap in space and time. The way that safety is guaranteed is by enclosing the UAS inside a bubble (with some given dimensions) that does not overlap with other bubbles. These bubbles set the separation minima between UASs. The goal is

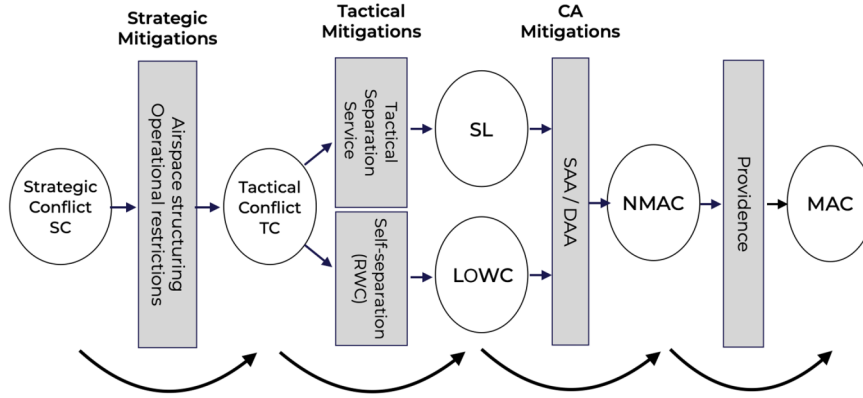


Fig. 1 Simplified version of SESAR barrier model.

to determine the separation distance required to achieve a given target level of safety (TLS) for UAS collisions. Determining a value for the separation distance implies replacing the qualitative model proposed by the SORA with a quantitative one. A quantitative model allows analysis of the impact of important factors, such as traffic density or navigation accuracy, on the separation minima.

The value of the separation minima in a self-separation scheme is also known as the remain well clear (RWC) distance. The term “well clear” was introduced by the International Civil Aviation Organization (ICAO) Rules of the Air [5] in the context of visual separation (self-separation), and it has no standardized value in manned aviation. However, the standardization of the RWC distance in UASs has attracted much interest, and some committees have presented detailed RWC proposals [6–9]. Nonetheless, the application of the RWC concept to unmanned aircraft is still the subject of some debate [10].

The problem first arose when airborne collision avoidance system (ACAS) Xu [11,12], which is a variant of ACAS X for unmanned aircraft initially intended for systems performing only the collision avoidance (CA) function, evolved into a full DAA system integrating both RWC and CA functions. Thus, the Radio Technical Commission for Aeronautics (RTCA) document SC-147, which is focused on the ACAS, was extended to RWC, which was the focus of RTCA document SC-228, working toward the development of minimum operational performance standards (MOPS) for UASs. The approach of SC-147 is quite different from that developed in SC-228: one of the hypotheses of the SC-228 work was that a UAS might fly without a CA function and rely solely on an RWC function. The definition and harmonization of the limits between RWC and CA became important issues when both committees started to work jointly. Moreover, the RWC distance, which was mainly coined for self-separation, requires some additional considerations to extend it to the separation provided by a TSS. This paper will use the term *separation minima* to refer to the separation provided by a TSS to disambiguate it from the RWC used in self-separation.

The research presented in this paper was performed as a part of the Building Basic Blocks for a U-Space Separation (BUBBLES) project [88], which is an exploratory research project within the Single European Sky Air Traffic Management (ATM) Research (SESAR) 2020 innovation pipeline. BUBBLES aims to provide a TSS in the context of U-space services. One of the key issues is defining the safety performance requirements and the required performance for the communication, navigation, and surveillance (CNS) systems supporting a TSS.

The remainder of this paper is organized as follows. Section II introduces the SESAR barrier model that defines the different phases and mitigations in air collisions. Section III presents a risk-based model for calculating the separation minima. Section IV presents an analysis of the influence of navigation performance on separation minima in UASs; then, Secs. V and VI, respectively, present detailed risk analysis of the strategic and tactical mitigations. Section VII

illustrates the previous analysis with a sample calculation of the separation minima in a given scenario. Finally, some conclusions are outlined in Sec. VIII.

II. Air Traffic Management Barrier Model

The BUBBLES risk model (see footnote §§) shown in Fig. 1 is a simplified version of the SESAR ATM barrier model for MACs [13]. The SESAR model sets different mitigations (so-called barriers) for preventing a strategic conflict (SC) ending in a MAC. Based on the SORA model [4], the BUBBLES model reduces the mitigations to the three big groups shown in the figure: strategic, tactical, and CA. In the timeline of a MAC, the SESAR model sets different barriers for preventing an SC successively degenerating into a tactical conflict (TC), a separation loss (SL) [or a loss of well clear (LOWC) in the case of self-separation], a near-midair collision (NMAC), or an actual MAC.

Strategic mitigations consist of procedures and operational restrictions adopted in the strategic phase (mission planning) that are intended to prevent SCs from becoming TCs by reducing the time of exposure between UASs. These include measures such as *operational restrictions* and *airspace structuring* (corridors, flight layers, etc.) aimed at avoiding the overlapping of UAS operations in time or space.

Tactical mitigations consist of aircraft maneuvers that are used to resolve TCs. These can be classified into two groups according to the *separator agent* that assumes the separation responsibility: in self-separation, each aircraft keeps its own separation and the pilot in command (PIC) is responsible; when using a TSS, the separation responsibility belongs to this service. The TSS is regarded as a centralized scheme, whereas self-separation is considered a distributed scheme. The reason for adopting a TSS is mostly the ability of this system to get a global picture of an airspace including the positions of all aircraft. This can be obtained using a surveillance system such as radar, and it enables the application of a more effective separation strategy. Several degrees of decentralization of this strategy are possible: from completely centralized to completely decentralized [14].

Finally, *CA mitigations* refer to last-resort mitigations, including actions triggered by visual detection and those provided by DAA devices. In the ACAS approach, a DAA system will have an RWC function and an optional CA function in a single device. In ACAS sXu (ACAS X for small unmanned aircraft) [11], these functions required redefinition. Rather than including separate RWC and CA functions, ACAS sXu includes one level of alerting and guidance, with the separation volume scaled based on intruder type [12,15].

III. Risk-Based Separation

The BUBBLES risk model is obtained by assigning a probability P_{TC} to TCs and a conditional probability to each barrier shown in Fig. 1. In this way, the probability of a MAC when using a TSS[¶] can be expressed as

^{§§}Vila Carbó, J. and Iocchi, L., “Algorithm for Separation Minima Determination,” BUBBLES—Defining the Building Basic Blocks for a U-Space Separation Management Service Project SESAR-ER4-31-2019, 2020.

[¶]A similar expression can be proposed for self-separation.

$$P_{\text{MAC}} = (1 - \text{Prov}) \cdot P_{\text{NMAC}|\text{SL}} \cdot P_{\text{SL}|\text{TC}} \cdot P_{\text{TC}} \quad (1)$$

where Prov stands for “providence,” which is the probability that an NMAC does not end in a MAC by the protective care of God; $P_{\text{NMAC}|\text{SL}}$ is the conditional probability that a MAC occurs after an SL; $P_{\text{SL}|\text{TC}}$ is the conditional probability that an SL occurs after a TC; and P_{TC} is the probability of a TC. In this expression, we implicitly assume that $P_{\text{NMAC}|\text{SL}} = 0$ and $P_{\text{SL}|\text{TC}} = 0$. Because modeling providence is unfeasible, a conservative worst-case approach is taken in which Prov = 0 will be assumed. In this way, Eq. (1) becomes

$$P_{\text{MAC}} \leq P_{\text{NMAC}|\text{SL}} \cdot P_{\text{SL}|\text{TC}} \cdot P_{\text{TC}} \quad (2)$$

The probabilities $P_{\text{SL}|\text{TC}}$ and $P_{\text{MAC}|\text{SL}}$ quantify the performance of the tactical and CA mitigations, respectively. These probabilities have been bounded by the American Society for Testing and Materials (ASTM) F3442/F3442M standard [16], which is a specification for DAA system performance requirements for relatively small UASs. The ASTM defines the risk ratio (RR) for an NMAC, denoted as RR(NMAC), as the ratio between the number of NMACs occurring with the mitigations provided by the DAA system and the number of NMACs occurring without mitigations; the LOWC risk ratio, denoted as RR(LOWC), is defined similarly. Note that the definitions of RR(NMAC) and RR(LOWC) match $P_{\text{NMAC}|\text{SL}}$ and $P_{\text{NMAC}|\text{LOWC}}$, respectively.

The performance requirements defined by the ASTM are listed in Table 1; DAA devices, such as ACAS sXu, are validated by proving conformance to these values [17]. This validation reveals an important fact: the risk ratio is a function of the alert response delay. The obtained RR value with automatic dependent surveillance–broadcast (ADS-B) out and the minimal delay in ACAS sXu is 0.2362 (40.95% better than that required by the ASTM). The reported values refer to self-separation, and so the risk values for separation with a TSS should improve them.

The rationale for risk-based separation calculations is to calculate the separation minima that will ensure that the P_{MAC} obtained using Eq. (1) meets an acceptable TLS. The TLS for air risk is usually expressed as the number of collisions per unit time: for example, 1×10^{-7} collisions per flight hour. According to this, Eq. (1) can be rewritten in terms of rates:

$$\mathcal{F}_{\text{MAC}} \leq P_{\text{MAC}|\text{SL}} \cdot P_{\text{SL}|\text{TC}} \cdot \mathcal{F}_{\text{TC}} \quad (3)$$

where $\mathcal{F}_{\text{TC}}$ is the conflict rate (conflicts per unit time), and $\mathcal{F}_{\text{MAC}}$ is the MAC rate (MACs per unit time). The terms $\mathcal{F}_{\text{TC}}$ and $P_{\text{SL}|\text{TC}}$ both depend on the separation minima (denoted as SEP), as will be shown. However, the term $P_{\text{MAC}|\text{SL}}$ depends on the CA distance rather than the SEP. This will be denoted as

$$\mathcal{F}_{\text{MAC}} \leq P_{\text{MAC}|\text{SL}} \cdot P_{\text{SL}|\text{TC}}(\text{SEP}) \cdot \mathcal{F}_{\text{TC}}(\text{SEP}) \quad (4)$$

The goal is to determine the separation value of SEP that meets $\mathcal{F}_{\text{MAC}} \leq \text{TLS}$.

The term $\mathcal{F}_{\text{TC}}$ can be minimized using strategic mitigations. According to the SORA [4], these mitigations are the major component of the risk, or “gross risk.” Tactical mitigations should face the remaining residual risk. A strategic mitigation such as an airspace structuring scheme that avoids spatial overlapping of operations will

bring $\mathcal{F}_{\text{TC}}$ close to zero, thus greatly reducing the performance required for tactical mitigations.

For convenience, the two conditional probabilities of Eq. (4) [$P_{\text{MAC}|\text{SL}}$ and $P_{\text{SL}|\text{TC}}(\text{SEP})$] can be combined into a single term $P_{\text{MAC}|\text{TC}}$ that represents the conditional probability that a TC becomes a MAC when both strategic and tactical mitigations fail:

$$\mathcal{F}_{\text{MAC}} = P_{\text{MAC}|\text{TC}}(\text{SEP}) \cdot \mathcal{F}_{\text{TC}}(\text{SEP}) \quad (5)$$

This rearrangement is used to highlight the fact that $\mathcal{F}_{\text{TC}}$ and $P_{\text{MAC}|\text{TC}}$ both depend on SEP but have opposite trends: $\mathcal{F}_{\text{TC}}$ increases with SEP, whereas $P_{\text{MAC}|\text{TC}}$ decreases with SEP. To understand this fact, consider a bubble of radius SEP around the aircraft. The probability that two bubbles overlap $\mathcal{F}_{\text{TC}}$ obviously increases with the bubble radius; however, $P_{\text{MAC}|\text{TC}}$ drastically decreases because the smaller the radius, the smaller the distance required to perform a separation maneuver. Using similar reasoning, it can be argued that term $P_{\text{MAC}|\text{SL}}$ in Eq. (4) depends on the radius of the CA volume rather than SEP. The determination of the CA volume goes beyond the scope of this paper.** The value of this probability will be assumed to be a requirement.

Another important fact is that $\mathcal{F}_{\text{TC}}$ depends on *traffic density*, whereas $P_{\text{MAC}|\text{TC}}$ does not. Clearly, $\mathcal{F}_{\text{TC}}$ depends on traffic density: the higher the traffic density, the greater the number of conflicts that will occur. However, $P_{\text{MAC}|\text{TC}}$ is assumed to be traffic-density independent because a traffic conflict usually has a pairwise nature: the probability that a third aircraft interferes in a traffic conflict resolution between two aircraft will be considered negligible for low/moderate traffic densities. Special care should be taken in conflict-resolution strategies to avoid cascade conflicts in conflict resolution. Under this assumption, the model for evaluating P_{TC} will be an all-versus-all model, whereas the calculation of $P_{\text{MAC}|\text{TC}}$ can be based on a pairwise conflict model. Finally, note that the model used for calculating $\mathcal{F}_{\text{TC}}$ is a model in which trajectories are not affected by conflict-resolution maneuvers. This is referred to as an *uncorrelated model* [18], whereas $P_{\text{MAC}|\text{TC}}$ must assume a *correlated model*.

As previously noted, the methodology presented in this paper for calculating the separation minima is based on establishing the dependence of $\mathcal{F}_{\text{TC}}$ and $P_{\text{SL}|\text{TC}}$ on SEP and determining the value of SEP that meets $\mathcal{F}_{\text{MAC}} \leq \text{TLS}$. The frequency $\mathcal{F}_{\text{TC}}$ will be calculated using some analytical or simulation model, as explained in Sec. V; and the probability $P_{\text{SL}|\text{TC}}$ will be shown in Sec. VI to depend on the time available to detect and solve a conflict, denoted as t_{sep} , and the effectiveness of the conflict-resolution strategies. The lower the required $P_{\text{SL}|\text{TC}}$ value, the higher the separation time t_{sep} needed to guarantee it. The values of SEP and t_{sep} are related through the *closure rate* of the two aircraft involved in an encounter, denoted as c :

$$\text{SEP} = c \cdot t_{\text{sep}} \quad (6)$$

The calculations for $\mathcal{F}_{\text{TC}}$ and $P_{\text{SL}|\text{TC}}$ are detailed in Secs. V and VI, respectively. Once these have been calculated, it must be verified that the resulting $\mathcal{F}_{\text{MAC}}$ is lower than the TLS. The TLS value that must be accomplished is discussed next.

The TLS sets an important upper bound on the desired level of risk. It is defined as the number of risk events per exposure time. In this way, the risk is expressed as a bound on the number of accidents, incidents, fatal injuries, money lost, etc., per flight hour. The risk analysis of this paper requires an upper bound to be set on the collision risk, expressed as collisions per flight hour. Therefore, if the proposed TLS is not given as a collision rate, some function will have to be developed to map it to this metric. This mapping is usually mission dependent. For example, if the TLS is given as the rate of fatal injuries on the ground caused by UAS collisions, this could be transformed into a collision rate by detailing ways in which collisions

**However, the CA volume can be determined following a similar methodology.

Table 1 ASTM performance requirements for DAA systems

DAA performance	RR(NMAC)	RR(LOWC)
Intruder equipped with transponder or ADS-B out	≤ 0.18	≤ 0.40
Noncooperative intruder	≤ 0.30	≤ 0.50

can cause fatal injuries. The population density in the flight area and the UAS's kinetic energy would have to be taken into account in these calculations.

The TLS for collisions between UASs and manned aircraft can use well-known standards for safety in manned aviation (summarized, for example, in Ref. [13]). In this way, SESAR sets a TLS of 1×10^{-9} collisions per flight hour for severity-1 incidents.^{††} However, some other criteria must be used to set the TLS for UAS–UAS collisions: the number of fatal injuries on the ground, the damage to ground infrastructure, the cost associated with the failure of the mission, the social impact, etc. Currently, there are no widely accepted TLS values for UASs. SORA v1.0 initially proposed a ground risk (UAS ground crashes) of 1×10^{-6} fatal injuries on the ground per flight hour. However, the TLS for air risk (UAS collision rate) is, in general, mission dependent.

A systematic way to determine the TLS for the collision rate of a mission could be assessing the impact of that rate on the TLS for all considered criteria for a given mission: fatal injuries, damage to infrastructure, and social impact. For example, a collision rate of 1×10^{-4} collisions per hour could be assessed to cause one fatal injury on the ground every 50 years, 25,000 euro losses per year, and two accidents (reported on social media) per year. If all these TLS values are acceptable, then the TLS for the collision rate can be accepted; otherwise, it will have to be set to a lower value.

IV. UAS Navigation Performance

UAS navigation performance is key for calculating separation minima. It strongly affects the assessment of the terms $\mathcal{F}_{TC}$, $P_{SL/TC}$, and $P_{MAC/SL}$ of Eq. (3), performed in the next sections. Therefore, this section will provide estimated values for the navigation performance of a UAS.

According to the *Performance-Based Navigation (PBN) Manual* [19], the overall positioning error is given by the total system error (TSE), which is the sum of the navigation system error (NSE), the flight technical error (FTE), and the path definition error (PDE). The NSE refers to the difference between the aircraft's position as estimated by the navigation system [e.g., global navigation satellite system (GNSS)] and its actual position. The FTE depends on the ability of the pilot or the flight control system (autopilot) to follow the defined path or track. The PDE is the difference between the defined path and the desired path, and this is usually negligible. All these errors are assumed to be Gaussian, zero mean, and independent; so, the total error distribution is also considered Gaussian with a 2σ (95th percentile) value $TSE = \sqrt{NSE^2 + FTE^2}$. Estimations for the NSE and TSE values in UASs are proposed next.

The estimation of the NSE is based on the recent International Organization for Standardization's (ISO's) 24355 proposal for the performance requirements of a GNSS for small civil and light multirotor UASs.^{†‡} This proposal estimates a value of 10 m for the horizontal error. The specification of Volume 10 of the ICAO's Annex 10 [20] for GNSS-based approaches in manned aircraft sets a requirement of 16 m, which is similar.

Vertical position errors based on basic GNSS (no augmentation) are estimated to be as high as 100 m in ISO standard 24355. Using real-time kinematics, the error can be reduced to 0.2 m. The GNSS is usually complemented in a UAS with barometric measures. Due to drifts and pressure changes, barometers are also quite error prone. The specification of Volume 1 of the ICAO's Annex 10 [20] for a GNSS approach with barometric vertical guidance (APV) sets this error in the range of 8–20 m, depending on the approach type (APV I or APV II). Due to the low performance of the barometric altimeters used in UASs, the vertical error will be estimated to a conservative value of 60 m.

The FTE is typically larger than the NSE. The required values of the FTE for manned aircraft are defined by ICAO document 9613: the *PBN Manual* [19]. They vary between 0.25 and 2.5 n miles, depending on the navigation specification and the phase of flight. Currently, there are no standard requirements for FTEs with UASs, but the DLR, German Aerospace Agency has performed some studies [21] to assimilate the performance of drones to PBN specifications. They have reported a nominal required navigation performance (RNP) value of approximately 10 m for 95% of the flight time. These values were obtained using different Global Positioning System receivers with a satellite-based augmentation system (SBAS). They concluded that this value could be used for the construction of drone-RNP airspace, but even this value could be too high for urban applications if no signal degradation is considered. However, they warned that a wider range of wind conditions would have to be investigated to confirm the observed TSE value.

The previous values refer to the horizontal accuracy of FTEs. Regarding vertical accuracy, PBN has not yet standardized any FTE value. However, the current proposals use a value of 50 ft (15.2 m) for GNSS+SBAS [22].

As previously noted, wind is an important factor that can contribute to significantly increasing the value of a FTE. The real problem with the wind in a UAS is wind shear: a difference in wind speed and direction over a relatively short distance. Uniform and constant winds have a smaller impact on the relative positions of two aircraft. Airplane pilots generally regard significant wind shear to be a horizontal change in airspeed of 30 kt for light aircraft and near 45 kt for airliners at flight altitude. Vertical speed changes greater than 4.9 kt also qualify as significant wind shear for aircraft [23]. The impact of wind on FTEs for different turbulence intensities has been assessed in a previous study [24]. The FTE of an aircraft with severe turbulence and a control system based on linear quadratic Gaussian and loop transfer recovery was found to have a mean value of 14.617 ft and an upper bound of 24.396 ft (4.5–7.4 m). The probability of severe turbulence is considered to be 1.36×10^{-5} . The proposal to account for windy conditions in this paper is to add these values to the TSE with no wind.

Based on previously introduced considerations, Table 2 summarizes the proposed 2σ values of the NSE, FTE, and TSE for a UAS in nonwindy and windy conditions.

V. Strategic Mitigations

This section describes ways to assess the effectiveness of strategic mitigations and the approaches for calculation of the term $\mathcal{F}_{TC}$ in Eq. (4), which represents the conflict rate. Conflict rates can be calculated using either *analytical models* or *simulation models*. Analytical models are probabilistic models such as the gas model or the Reich–Marks model [25]. These models have been used by the ICAO in document 9689 [26] for the determination of current separation minima in manned aviation. Simulation models include Bayesian networks based on radar data [18] and Monte Carlo simulations. In this work, simulation models were used to estimate conflict rates in particular scenarios defined by specific airspace structuring.

A. UAS Mission Example

Consider a mission in which there is a fleet of UASs performing surveillance of an area close to the border between Spain and France. Figure 2 shows the horizontal profile for the simulated trajectories of the mission; Fig. 3 shows the vertical profile, in which noisy trajectories are observed because of the assumed positioning error.

The mission consists of five UASs flying in scan patterns embedded within an operational area spanning a rectangular surface

Table 2 Proposed navigation performance summary for UAS

	NSE, m	FTE		TSE	
		No wind	Wind	No wind	Wind
Lateral	16	10.0	17.4	18.9	23.7
Vertical	60	15.2	22.6	61.9	64.1

^{††}Incidents in which an aircraft comes into physical contact with another aircraft in the air.

^{†‡}General Requirement of Flight Control System for Civil Small and Light Multirotor UAS," International Organization for Standardization Standard Draft TC 20/SC 16/WG 6, 2020.

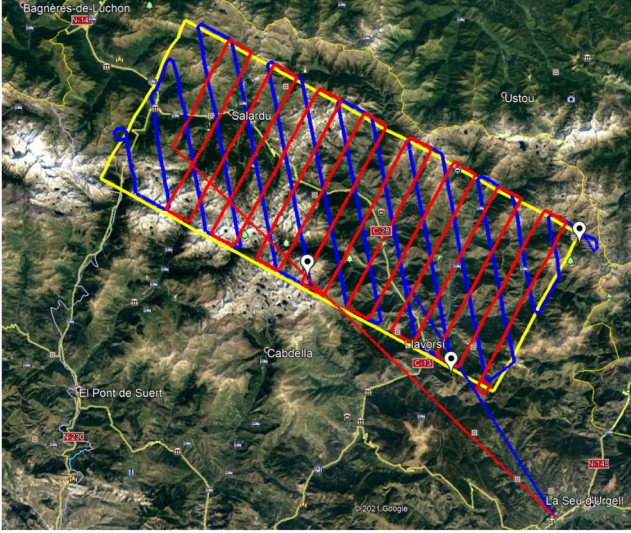


Fig. 2 Border surveillance mission: horizontal profile.

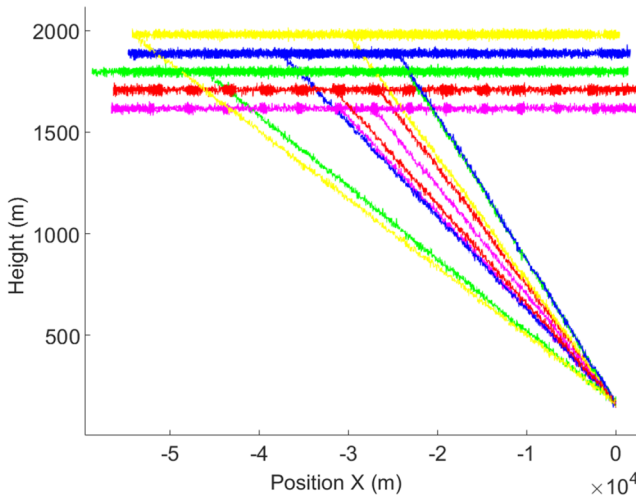


Fig. 3 Border surveillance mission: vertical profile.

of 50×20 km (1000 km^2). The airspace is assumed to be segregated, and no intruders can enter the operations volume. The patterns have different distances between their scan legs and different azimuth angles. The UASs are based at the aerodrome of La Seu d'Urgell. An access corridor from this aerodrome to the operations area is enabled. The UASs selected for this mission have a nominal speed of 80 kt.

The basic strategic mitigation proposed in BUBBLES to avoid conflicts between UASs is to structure the airspace into vertical layers and allocate each UAS to a different layer. According to this scheme, the described mission assumes that each of the five UASs is allocated a different flight layer and is not expected to perform flight-layer changes during its mission other than to avoid conflicts or collisions. A nominal separation SEP_V between flight layers is set, as shown in Fig. 4. Despite the vertical separation, traffic conflicts may occur because UASs have vertical position errors, which are assumed to be Gaussian. The protection volume that delimits the conflict threshold is a cylinder of radius λ_{xy} and height $2\lambda_z$. The nominal separation between the altitude levels is $SEP_V = 2 \times \lambda_z$.

According to the Reich model [25], the conflict rate for two aircraft with the previously defined protection volume is

$$\mathcal{F}_{TC} = \mathcal{F}_H \cdot P_{TC|H} \quad (7)$$

where $\mathcal{F}_H$ is the frequency of a horizontal overlap between two UASs, referring to an intruder UAS entering an infinite cylinder of radius λ_{xy} centered on the host aircraft; and $P_{TC|H}$ is the conditional

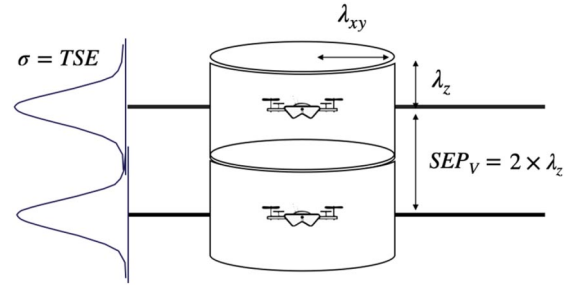


Fig. 4 Model for vertical separation of UASs.

probability that an intruder in an infinite cylinder of radius λ_{xy} enters the cylinder of height $2\lambda_z$.

B. Simulation Model for Conflict Frequencies

The accuracy of the simulation models strongly relies on their ability to generate representative traffic patterns in the considered scenarios. The sample mission was simulated using the BB-Planner tool developed in the BUBBLES project. This tool was specifically designed to create mission plans and simulate UAS missions. Unlike other tools used in manned aviation, BB-Planner allows UAS trajectories such as scan patterns, delivery missions, and missions between a network of destinations to be easily defined. It also allows specification of traffic densities, positioning errors, and UAS performance parameters.

The goal of the simulations was to analyze the conflict rate $\mathcal{F}_{TC}$ as a function of the horizontal λ_{xy} and vertical λ_z separations. A Monte Carlo simulation with 1000 trajectory samples for each value of λ_{xy} and λ_z was performed. The UAS navigation performance in the simulations was characterized by the previously estimated values of TSEs. Trajectory samples were generated by randomizing some of their characteristics (scan pattern angle and scan pattern distance) and departure times.

Figure 5 shows that the conflict rate increases with horizontal separation λ_{xy} for all values of the vertical separation λ_z . This is because horizontal overlaps $\mathcal{F}_H$ increase (almost linearly) with λ_{xy} . The conflict also increases strongly with λ_z . For example, it can be seen that the conflict rate for $\lambda_{xy} = 1500$ and $\lambda_z = 50$ is close to 0.8 conflicts per hour. This frequency is a little smaller than the theoretical results that were obtained using the gas model of 2.5 conflicts per hour, although they are the same order of magnitude. The difference is due to the fact that the gas model uses random trajectories and does not take trajectory patterns into account.

The values obtained for $\mathcal{F}_{TC}$ allow important decisions to be made about strategic and tactical mitigations. If $\mathcal{F}_{TC}$ is greater than the

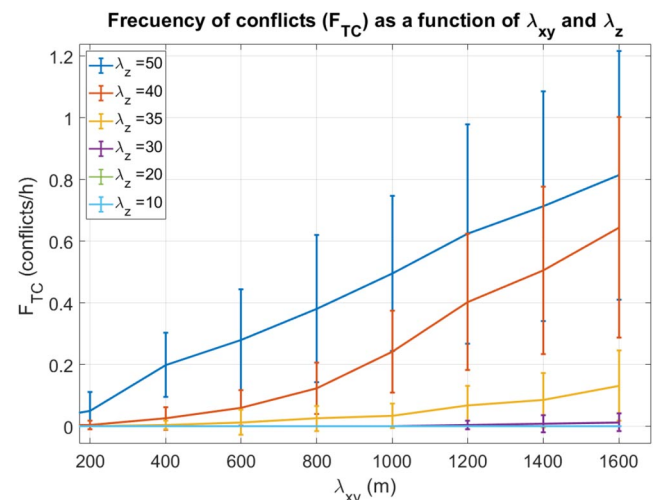


Fig. 5 Simulation results for the border surveillance mission showing the variations of the conflict rate with λ_{xy} and λ_z .

conflicts that can be solved using any tactical deconfliction service (for example, ≥ 200 conflicts per hour), then either the traffic capacity of a sector has to be reduced or a better strategic deconfliction scheme must be adopted. The value of $\mathcal{F}_{TC}$ can also be used to decide whether to use a self-separation scheme or a ground-based TSS: a low conflict rate would make TSS unnecessary, and so self-separation would be the right choice.

VI. Tactical Mitigations

Tactical mitigations are maneuvers performed within a timeframe nominally sufficient to avoid an SL. From our perspective, tactical mitigations are used to minimize the value of $P_{SL|TC}$ in Eq. (4). The criteria for setting a “nominally sufficient timeframe” for defining the separation minima are quantified in the method described in the following:

The proposed method for estimating the separation minima is the *encounter timeline* concept, which has also been used in previous works [27–29]. This concept defines the different events involved in the separation process and the allocation of latencies to each event.

The events in the encounter timeline are represented in Fig. 6; these are common to both self-separation and ground-based separation. The time allocated to each event depends on the separation mode and the automation level used for each event, from human-performed to fully automatic executions [30]. The encounter timeline latencies are described as follows:

1) *Surveillance* includes the latencies of two tasks: aircraft sensing and tracking. *Aircraft sensing* is the detection of the positions of the aircraft involved in an encounter; *tracking* is the processing of the aircraft’s instantaneous positions to provide a set of highly accurate tracks.

2) *Separation* includes the latencies of two tasks: conflict detection and conflict resolution. *Conflict detection* involves the application of some criteria to every pair of detected aircraft to determine whether an SL is close to occurring; *conflict resolution* is a decision-making process for selecting the appropriate maneuver to prevent an SL.

3) *Communication* involves the transmission latency between the entity that provides the separation service and the pilot. If communication is not fully reliable, the communication latency should also account for message retransmissions. This latency is considered null for the case of self-separation.

4) *Piloting* represents the time required for the pilot to cognitively process a clearance, make a decision, and execute it. Different automation levels can be considered for this event, leading to different latencies.

5) *Aircraft maneuver* is the time required for the aircraft to achieve a change of trajectory sufficient to reach a clear-of-conflict state. Several separation maneuvers can be used: horizontal, vertical, and hovering (in the case of rotary-wing UASs).

The proposed method for calculating the separation minima uses the same criteria as that of ICAO documents 4444 [31] and 9869 [32] for manned aviation. The allocation of times to CNS in these documents is based on three performance-based parameters (the required communication performance, the RNP, and the required surveillance performance) specified in RTCA standard DO-306 [33] and European Organisation for Civil Aviation Equipment (EUROCAE) standard ED-122 [34]. However, the proposed values

are not directly applicable to UASs because their CNS infrastructures are very different.

The time allocated to each event of the encounter timeline will be characterized as a probabilistic distribution. The overall time of the encounter timeline will be also characterized as a probabilistic distribution that is a combination of all the probabilistic distributions of each stage therein.

The time distribution of the encounter timeline establishes the relationship between the separation time/distance and the probability $P_{SL|TC}$ that a TC becomes an SL. However, $P_{SL|TC}$ also depends on the effectiveness of tactical mitigations, denoted by ϵ_{tac} . This represents the probability that a conflict can be effectively detected and solved, provided that tactical mitigations are activated in a timely manner. In summary, $P_{SL|TC}$ can be expressed as

$$P_{SL|TC} = F(t) \cdot \epsilon_{tac} \quad (8)$$

where $F(t)$ is the cumulative distribution function that provides the likelihood that a conflict can be detected and solved in a time less than or equal to t [$F(t) = P(T \leq t)$]. The next subsections analyze the probabilistic time allocated to each step of the encounter timeline.

A. Surveillance Time

The surveillance delay includes the aircraft sensor latency, track processing time, and latency of the surveillance network. A representative estimation for the surveillance delay could be the ACAS sXu MOPS [8,9] requirements for DAA systems. ACAS sXu sets a requirement for a total surveillance latency of 3.5 s based on the fact that surveillance is implemented as a periodic task with a 1 Hz execution rate (1 s). However, part of this latency can be compensated for by extrapolation, yielding a 0.3–0.7 s uncompensated latency.

If the surveillance and separation processes are performed at the same site as a unique service, the network latency between those tasks will be negligible. This is the case with a ground-based TSS. Otherwise, the latency of the surveillance network should be included. This would be the case for a self-separation scheme in which surveillance relies on a ground-based service and the separation process is performed on board.

The uncompensated latency of the surveillance task will be increased in a conservative way from the value of ACAS sXu to 2 s to account for the latency of the surveillance network and the computational overload of track processing. The estimated time for surveillance delay in a UAS environment is much lower than the requirement in RTCA standard DO-306 [33] and EUROCAE standard ED-122 [34] of 90 s for manned aviation under normal conditions. This large time requirement is due to the fact that ADS-B is not available in oceanic airspace.

B. Separation Time

The execution times and effectiveness of the steps involved in the separation process are highly dependent on the automation level: human-in-the-loop systems are usually more reliable, but machine-based systems are generally faster. There is usually a tradeoff between processing time and effectiveness.

The ACAS sXu MOPS specifies an alert generation and execution time of 2.8–6.8 s for a DAA system. This includes not only the ACAS

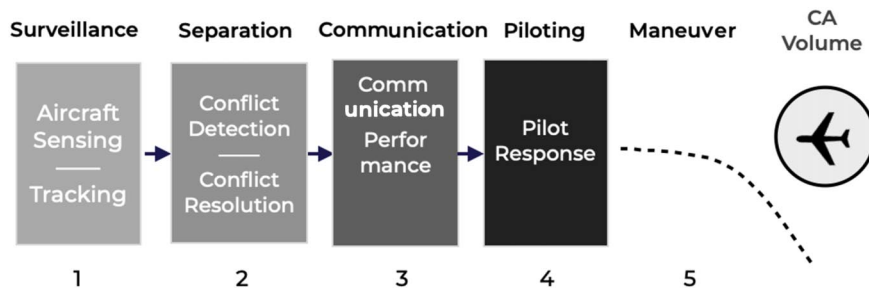


Fig. 6 Encounter timeline concept.

sXu conflict detection and resolution times but also the communication and piloting delays. Extracting the conflict detection and resolution time from this total time is difficult. However, the experiments of Ref. [35] revealed that automatic conflict detection and resolution require extremely small times using “dynamic programming”: less than 4 ms for 16 intruders on a single-core processor.

To estimate the conflict and resolution times with a human-in-the-loop system, some studies of human factors have been reviewed. The experiments of Ref. [36] measured the response times of ATC ground controllers. The results showed that the mean response time to an alert was 4.6 s with a maximum of 8.1 s. The mean response duration was 2.3 s with a maximum of 5.3 s. Considering the overall delay (reaction plus response duration) and taking maximum times as a 2σ value, these values can be mapped to a normal distribution with $\mu = 6.9$ s and $2\sigma = 4.9$ s. RTCA standard DO-306 [33] and EUROCAE standard ED-122 [34] set a much higher delay bound of 30 s for manned aviation.

The applicability of ACAS sXu MOPS times to a ground-based TSS is questionable: the TSS does not only solve last-resort conflicts but also the separation between several aircraft on a longer time horizon. However, a TSS for UASs is supposed to use a higher level of automation, although no prototype providing results regarding times has been reported yet. For this reason, the separation time will be estimated in a conservative way, using the values previously reported for ATC controllers in Ref. [36].

The separation process must be characterized not only for its delay but also for the effectiveness of its tactical mitigations. This effectiveness is bounded by the requirements of the ASTM [16] for the RR of the loss of well clear. This RR is defined as the number of well-clear losses with mitigations divided by the number of well-clear losses without mitigations. The ASTM requires this RR to be $\leq 40\%$ for intruders equipped with cooperative surveillance systems (ADS-B out) and $\leq 50\%$ for noncooperative intruders. In this way, our estimation for the effectiveness of tactical mitigations will be set to $\geq 60\%$.

C. Communication Time

The communication time is the delay estimated between the entity providing separation and the PIC. For the case of self-separation, the separation process is assumed to be performed on board and the communication delay is negligible. In a ground-based TSS, the communication network latency must be estimated.

For the latency estimation, the network will be assumed to not be completely reliable, and so messages can get garbled or lost. Although the propagation times of most networks are negligible, to maintain reasonable bandwidth utilization in a broadcast channel, periodic scheduling of messages is recommended: for example, at a 1 Hz frequency. It will be reasonably assumed that 68% of the messages get through in the nominal period of 1 s, whereas lost messages will need two or more periods. In this way, the estimation for the communication time will be approximated to a Gaussian distribution with $\mu = 1$ s and $\sigma = 2$ s.

D. Piloting Time

Pilot response times depend on human factors, and so they are highly variable. In ACAS sXu MOPS, these times are included in the alert generation and execution time of 2.8–6.8 s reported in the previous section. Additional sources for estimating UAS piloting times are experiments performed by NASA [37]. They reported several measured response times of UAS pilots after an ATC clearance. The one considered in this paper is the time between when the air traffic controller finished issuing the command to when the

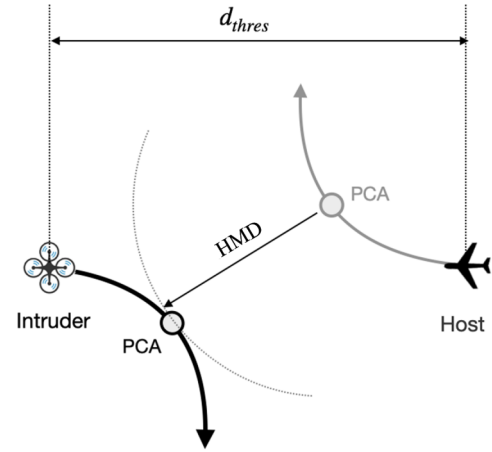


Fig. 7 Required horizontal distance for separation maneuver.

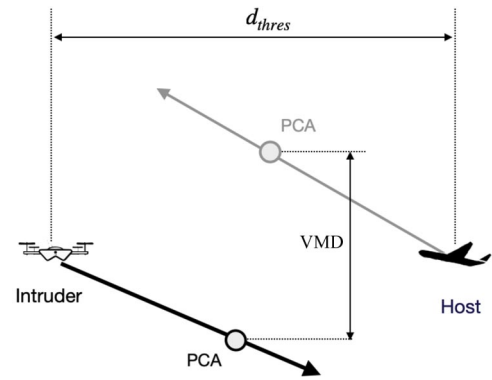


Fig. 8 Required vertical distance for separation maneuver.

pilot began to execute it. Different ATC commands were considered in the experiments, and the results for each command are summarized in Table 3.

For the estimation of the piloting time for a separation maneuver, it will be assimilated to a “route amend heading/altitude” or to a “traffic alert” + “immediate turn” command. The most conservative time estimations for this are $\mu = 4.77$ s and $\sigma = 3.11$ s. The value set by the RTCA standard DO-306 [33] and EUROCAE standard ED-122 [34] for manned aviation is 30 s, which (as in previous estimations) is considerably larger.

E. Aircraft Maneuver Time

The last step in the encounter timeline is the separation maneuver. The goal is to estimate the distance d_{thres} at which a separation maneuver between two aircraft should begin to guarantee a minimum miss distance between them at the point of closest approach (PCA). This is represented in Fig. 7 in the horizontal plane and in Fig. 8 in the vertical plane. Here, HMD is the horizontal miss distance between two aircraft at the PCA, and VMD is the vertical miss distance. The goal is to calculate the distance d_{thres} , as shown in Figs. 7 and 8, at which the aircraft should initiate the separation maneuver to keep the HMD (or VMD) at the PCA. This first requires estimation of the HMD and VMD. This estimation will be based on the *aircraft envelope* (AE) concept.

Table 3 Pilot response times for UAS maneuvers in seconds

	Crossing restriction	Direct to	Route amend-altitude + traffic	Route amend heading	Route amend altitude	Traffic alert + immediate turn
μ (mean)	7.63	7.32	1.88	4.77	2.63	1.70
σ (standard deviation)	5.66	7.14	2.17	3.11	1.91	0.93

The AE concept is a generalization of the CA volume concept. The CA volume, as defined in Refs. [11,38], is a cylinder with a radius of 500 ft (152.4 m) and a half-height of 100 ft (30.48 m), which roughly corresponds to double the dimensions of the largest civil transport aircraft. The AE is defined as the CA volume extended by two distances: 1) the position uncertainty due to the TSE, and 2) the stationary motion (SM) boundary.

In this way, the cylinder defined by the AE has the following dimensions:

$$\text{RAE} = R_{\text{size}} + \text{TSE}_h + \text{RSM} \quad (9)$$

$$\text{HAE} = H_{\text{size}} + \text{TSE}_v + \text{HSM} \quad (10)$$

where RAE is the radius of the aircraft envelope, R_{size} is the radius of the aircraft perimeter in the horizontal plane, TSE_h is the total horizontal system error, and RSM is the radius of the SM boundary as defined the following. The corresponding magnitudes for the vertical plane are H_{size} , TSE_v , and HSM, respectively. Once the RAE for a pair of conflicting aircraft has been calculated, then the HMD can be found as the sum of both; the process is similar for the VMD.

The SM boundary is the envelope that encloses the aircraft position for a given period of time. For example, if a UAS is hovering over some point, as shown in Fig. 9, the SM boundary in the horizontal plane is a circle that encloses all possible positions of the aircraft trajectory over that time. For the purpose of calculating the separation distance, the host aircraft is considered as a “big

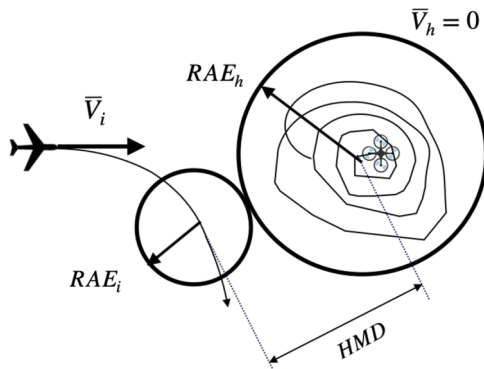


Fig. 9 AE concept applied to hovering.

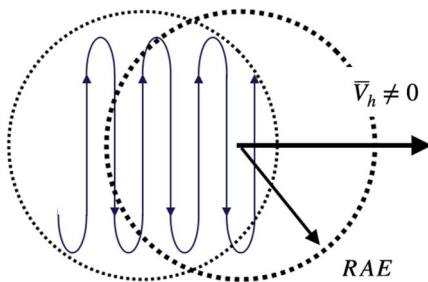


Fig. 10 AE concept applied to a scan pattern.

point” of the RAE with an average speed of $\bar{V}_h = 0$; i.e., the host is still.

The average speed of the AE, denoted as $\bar{V}$, is defined as the average of the instantaneous speed vectors of the aircraft over the last k instants of time, where k is the length of the time period:

$$\bar{V}(t) = \frac{1}{k} \sum_{t-k}^t V(t) \quad (11)$$

If an aircraft is flying in a straight trajectory with constant speed V , then it has a small envelope ($R_{\text{SM}} = 0$) because the position variance is maximum (not stationary) and the average speed $\bar{V}$ is equal to the instantaneous speed V . In a scan pattern (see Fig. 10), $R_{\text{SM}} > 0$ and $\bar{V} > 0$. In this case, the average speed $\bar{V}$ is the translational speed of the UAS along the longitudinal axis of the scan pattern.

Once the HMD and VMD have been determined, d_{thres} can be calculated by trajectory simulation of the separation maneuvers in the horizontal and vertical planes shown in Figs. 7 and 8 using a simple flight-dynamics model with three degrees of freedom (translational motion). The simulated trajectories of the separation maneuvers depend on the performance and maneuverability parameters of the UAS. Although these vary, the RTCA standard DO-365B [7] and EUROCAE ED-275B [9] MOPS for DAA systems set some “standard” maneuverability parameters that are used in UAS avoidance maneuvers, such as the latency, turn rate, and acceleration.

The distance d_{thres} depends much on the encounter geometry, the type of UAS trajectory (linear, hovering, scanning, etc.), and the closure rate of the aircraft involved in an encounter. The simulations deal with these aspects using the AE concept: the CA volume is replaced with the AE, and the instantaneous speed of the aircraft is replaced with the average speed. Additionally, speeds depend much on the aircraft type, and so the simulations consider encounters between three types of aircraft: a small UAS (SUAS), a large UAS (LUAS), and a typical manned aircraft flying in VFR airspace. The assumed speeds for these aircraft are 13, 20, and 52 m/s, respectively.

The simulations estimate the AE by adding the size of the considered aircraft; a TSE value that corresponds to wind turbulence (worst-case TSE); and the RSM, which is zero for linear trajectories. The HMD value at the PCA must be equal to the sum of the AEs of the host and intruder.

Table 4 shows the simulation results for the time/distance ($t_{\text{thres}}/d_{\text{thres}}$) estimations for both the horizontal and vertical separation maneuvers. All values are significantly smaller than the conservative estimation of 75 s required by RTCA standard DO-306 for manned aircraft [33].

The results for d_{thres} in the horizontal maneuver depend on the encounter angle between the host and the intruder. The worst case is 180 deg (head-on), and the best-case is 0 deg (from behind). The result is a uniform distribution of times/distances indicated as an interval. For the SUAS, the time distribution is [0.4, 8.1] s. The results in time are slightly worse for the SUAS because it is slower to maneuver, but the distances are higher for the LUAS.

The vertical maneuver takes much longer than the horizontal one: [16.3, 17.9] s vs [465, 651] m. The table only shows the value of 17.9 that corresponds to head-on encounters in UAS corridors because with a slight trajectory misalignment in a head-on encounter, the

Table 4 Time/distance estimations for the separation maneuver

Encounter	Closure rate, m/s	HMD, m	VMD, m	Horizontal maneuver		Vertical maneuver	
				t_{thres}	d_{thres}	t_{thres}	d_{thres}
SUAS–SUAS	26	48.4	129.2	[0.4, 8.1]	[49, 221]	17.9	465
LUAS–LUAS	40	49.4	130.2	[0.6, 6.7]	[50, 274]	16.3	651
SUAS–VFR	65	57.9	130.7	[13.1, 13.2]	[533, 846]	28.1	1828
LUAS–VFR	72	58.4	131.2	[10.4, 10.6]	[358, 754]	28.0	2018

intruder moves out of the AE in the horizontal plane quickly, and horizontal separation holds before vertical separation is eventually accomplished. Large values of separation are due to the large assumed VMD value of about 130 m, reflecting the low confidence for UAS altimeter accuracy. A VMD value of 50 m (similar to the SBAS required accuracy) would reduce the time from 17.9 to 6.7 s.

The simulations of encounters between a UAS and a VFR aircraft assume that the VFR aircraft is not aware of the UAS, and so it does not perform any separation maneuvers; only the UAS undertakes collision avoidance. This leads to higher separations: the horizontal maneuver requires a value of t_{thres} close to 13 s for the SUAS and 10.5 s for the LUAS. The results for the vertical maneuver are close to 28 s in all cases.

F. Total Time

The overall time for the encounter timeline is the combination of the probabilistic distributions of all the stages. Table 5 summarizes the times for all the timeline stages except the aircraft maneuver. Most times are characterized as Gaussian distributions. The exceptions are the surveillance time, which is considered deterministic, and the aircraft maneuver time, which is modeled as a uniform distribution over the range of encounter angles. All of the distributions except for the aircraft maneuver are also considered to be Gaussians, with means and standard deviations as shown in Table 5.

The sum of these Gaussian distributions and the uniform distribution of the aircraft maneuver has a density function that can be easily proven to be

$$F(t) = \frac{1}{(b-a)} [\Phi_{\mu,\sigma}(t-a) - \Phi_{\mu,\sigma}(t-b)] \quad (12)$$

where a and b are the parameters of the uniform distribution, and Φ is the cumulative distribution function of the resulting Gaussian distribution. The results for the horizontal separation distance SEP for encounters between different types of aircraft obtained with the resulting distribution are shown in Table 6 for different confidence intervals.

It can be seen that they are slightly higher than the 2000 ft (610 m) of the Sense and Avoid Science and Research Panel (SARP) recommendation for RWC distance in UAS-UAS encounters. The values for UAS-VFR encounters are significantly higher, as might be expected. These values make sense when compared to the SARP values because, unlike this study, the SARP assumes a self-separation scheme with lower conflict detection times and communication overhead. If the conflict detection was reduced from 6.9 to 1.0 s and the communication time was considered negligible, the experiment would yield a value of 652 m, which is very close to the SARP recommendation. The vertical separation is given by the minimum VMD required in the PCA, which is close to 130 m. This almost doubles the SARP recommendation

value of 250 ft (76.2 m). This is mainly due to the conservative assumption for the baroaltimeter accuracy.

VII. Separation Calculation Example

This section illustrates the calculation of the separation between two SUASs in a simple scenario like the one described in Sec. V.A. This is a risk-based calculation that is performed using Eq. (4):

$$\mathcal{F}_{\text{MAC}} = P_{\text{MAC|SL}} \cdot P_{\text{SL|TC}}(\text{SEP}) \cdot \mathcal{F}_{\text{TC}}(\text{SEP}) \leq \text{TLS}$$

The terms $P_{\text{SL|TC}}(\text{SEP})$ and $\mathcal{F}_{\text{TC}}(\text{SEP})$ both depend on horizontal separation minima, whereas $P_{\text{MAC|SL}}$ depends on the effectiveness of CA mitigations. Determining the separation minima involves determining the minimum SEP that meets this equation.

The first step is selecting a TLS. The TLS for fatal injuries is assumed to be less than one fatal injury in 50 years. Assuming 1000 flight hours per year, this yields a TLS of 2.0×10^{-5} fatal injuries per flight hour. Only 10% of the fatal injuries are assumed to be due to UAS collisions. The TLS for damage to ground infrastructure is assumed to be less than 10,000 per year. The goal is to obtain the overall TLS, in collisions per hour, from the previous values. Although this calculation exceeds the scope and length of this paper, let us assume that the TLS obtained with these values for $\mathcal{F}_{\text{MAC}}$ is 4.0×10^{-4} collisions per hour.

The second step is determining the time t that the encounter timeline requires for different probabilities of $F(t)$. This is performed by numerically solving Eq. (12). Table 7 shows the probabilities for $F(t)$ and the corresponding SEP values as times and distances.

The third step is calculating $P_{\text{SL|TC}}$ as a function of SEP. This is achieved using a variation of Eq. (8): $P_{\text{SL|TC}} = (1 - F(t)) \cdot (1 - \epsilon_{\text{tac}})$. It is assumed that $\epsilon_{\text{tac}} = 1 - \text{RR}(\text{LOWC})$ with $\text{RR}(\text{LOWC}) = 0.4$, which is the minimum performance defined by the ASTM for the $\text{RR}(\text{LOWC})$ in DAA systems. The results are shown in Table 7.

The fourth step is calculating $P_{\text{MAC|SL}}$, which is the risk ratio of DAA systems. This value is based on the ASTM [16] values for $\text{RR}(\text{NMAC})$ equipped with cooperative surveillance systems (ADS-B out). The results are shown in Table 7.

The fifth step is calculating $\mathcal{F}_{\text{TC}}$ as a function of SEP, either using the simulations of Sec. V or the gas model. Assuming the performance parameters of an SUAS, a vertical separation of $\text{SEP}_V = 100$ m, and values of SEP corresponding to the different confidence intervals in Table 7, the obtained values of $\mathcal{F}_{\text{TC}}$ are shown in the row for $\mathcal{F}_{\text{TC}}$. The results show a conflict rate between 0.25 and 0.60 conflicts per hour. These values can be used to select the type of tactical separation used for a given airspace. The small values of $\mathcal{F}_{\text{TC}}$ obtained for the different separations in this case mean that a ground-based TSS would have low utilization, and self-separation could be adequate.

The resulting MAC rates $\mathcal{F}_{\text{MAC}}$ are obtained using Eq. (4). The results are shown in the last row of Table 7. It can be seen that for a required TLS of 4×10^{-4} collisions per hour, the required separation threshold is between 30.0 and 37.0 s. The separation obtained by iterating numerically is approximately 32.5 s, or 845 m. The AE concept is not applied here because the envelopes of the different aircraft are completely overlapping and cannot be considered as big points from the other aircraft. It could be applied if the intruder was

Table 5 Encounter timeline distribution times

Encounter timeline stage	μ , s	σ , s
Surveillance service	2	0
Separation service	6.9	4.9
Communication	1	2
Piloting	4.8	3.1
Total	14.7	6.1

Table 6 Time/distance estimations for the separation minima

Encounter	Closure rate, m/s	T , s			SEP, m		
		95%	99.7%	99.99%	95%	99.7%	99.99%
SUAS-SUAS	26	29.7	37.0	62.6	772	962	1628
LUAS-LUAS	40	28.9	36.0	61.4	1156	1440	2456
SUAS-VFR	65	37.9	44.7	50.7	2464	2906	3296
LUAS-VFR	72	35.3	42.1	48.0	2542	3031	3456

Table 7 Relationship between SEP, $\mathcal{F}_{\text{TC}}$, and $\mathcal{F}_{\text{MAC}}$

	Probabilities for $F(t)$			
	86.7%	95.5%	99.7%	99.99%
SEP, s/m	26.2/681	30.0/780	37.0/962	62.6/1628
$P_{\text{SL TC}}$	5.3×10^{-2}	1.8×10^{-2}	1.2×10^{-3}	4.0×10^{-5}
$P_{\text{MAC SL}}$	1.8×10^{-1}	1.8×10^{-1}	1.8×10^{-1}	1.8×10^{-1}
$\mathcal{F}_{\text{TC}}$, conflict/h	2.5×10^{-1}	2.9×10^{-1}	3.6×10^{-1}	6.0×10^{-1}
$\mathcal{F}_{\text{MAC}}$	2.4×10^{-3}	9.4×10^{-4}	7.8×10^{-5}	4.3×10^{-6}

flying in a straight trajectory penetrating the operation volume of the UAS.

VIII. Conclusions

This paper has presented a risk-based method for determining the separation minima. This method shows how a TLS for UAS collisions can be accomplished as a tradeoff between strategic and tactical mitigations. Strategic mitigations mostly depend on traffic density and traffic structuring, whereas tactical mitigations depend on the separation time and the effectiveness of tactical mitigations. The separation time is calculated using the encounter timeline concept, which is structured as a number of stages between aircraft conflict and collision. The time and effectiveness associated with each stage is highly dependent on the separation scheme (self-separation or ground-based TSS) and the automation level. The impact of UAS-specific trajectories (not straight lines, as in manned aviation) on the separation minima can be taken into account using the AE concept, which is a generalization of the CA volume. The resulting separation distances show that the proposed method provides separations that are slightly higher than that proposed by the ASTM for a self-separation scheme.

Finally, it can be stated that having unique separation minima in UASs, in a similar way to manned aviation, is difficult because of the diversity of UAS characteristics, performance, operations, and trajectory types. However, in this work, the different factors upon which the separation minima depend were analyzed, and this allowed quantitative risk analysis to be performed and pairwise separation minima to be estimated for a given case. Providing pairwise separations implies using a high automation level.

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